



Qualification Pack

Machine Operator - Injection Molding

QP Code: CPC/CAP/Q0201

Version: 1.0

NSQF Level: 3.5

CENTRAL INSTITUTE OF PETROCHEMICALS ENGINEERING & TECHNOLOGY || CIPET Head Office, T.V.K.
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CPC/CAP/Q0201: Machine Operator - Injection Molding

Brief Job Description

The Machine operator Injection molding understands the instructions from Supervisor for completing the work order and helps in handling plastics granules (raw material), operates the Plastic Injection Moulding machines & Quality and Inspection of product as per the requirement

Personal Attributes

Technical aptitude Safety-conscious Reliability and punctuality Organizational skills Problem-solving ability Teamwork and communication Quality-focused mindset Willingness to learn

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [CPC/CAP/N0201: Basic concept, job requirements and basics know how related to the process](#)
2. [CPC/CAP/N0202: Injection moulding related operations, monitor process parameters and troubleshoot the process/product if any](#)
3. [CPC/CAP/N0203: Conduct quality checks and inspection of the finished products with reference to the approved product](#)
4. [CPC/CAP/N0204: Entrepreneurship in Injection Moulding](#)
5. [CPC/CAP/N0401: Basic health and safety practices at the workplace, 5S](#)
6. [CPC/CAP/N0402: Basics of computer and data entry in MS OFFICE/office Open source suite Software](#)
7. [DGT/VSQ/N0101: Employability Skills \(30 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Chemical & PetroChemical
Sub-Sector	Petrochemicals
Occupation	Injection Moulding
Country	India

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NSQF Level	3.5
Credits	20
Aligned to NCO/ISCO/ISIC Code	NCO-2015/8142.0501
Minimum Educational Qualification & Experience	11th Class with NA of experience OR 10th grade pass with 1.5 years of experience OR 8th Class pass with 4.5 years of experience OR Previous relevant Qualification of NSQF Level with 1.5 years of experience
Minimum Level of Education for Training in School	
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	25/05/2028
NSQC Approval Date	26/05/2025
Version	1.0
Reference code on NQR	QG-3.5-CP-04117-2025-V2-CIPET
NQR Version	2

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CPC/CAP/N0201: Basic concept, job requirements and basics know how related to the process

Description

This NOS unit is about understanding the job requirement, operating the injection moulding machine in order to produce good quality plastic parts using given material and mould.

Scope

The scope covers the following :

- This unit/ task covers the following:
- Understanding the work order and the process requirement from the supervisor
- Arranging the required raw material and Moulds for the process
- Cleaning the equipment and the moulds

Elements and Performance Criteria

Understand the work and the process requirements

To be competent, the user/individual on the job must be able to:

- PC1.** To interact with the supervisor in order to understand the production schedule
- PC2.** To plan the day's production activities based on the supervisor's instructions
- PC3.** To ensure availability of consumables and plastics materials for production in sufficient quantity as per production plan/supervisor instructions
- PC4.** Clearly understanding the does and don'ts of the manufacturing process as defined in SOPs/ Work Instructions or defined by supervisors
- PC5.** Check availability of the personal protective equipments (PPE) like Gloves, Goggles etc.
- PC6.** Understand the molding procedure and process to be adopted for completing the work order from the supervisor by referring the Work Instruction document/ SOP manual.

Arrange for the material to be moulded and Mould required for the same

To be competent, the user/individual on the job must be able to:

- PC7.** Ensure that the required material is procured from the store before starting the process
- PC8.** Understand the Mould required for executing the required operation and ensure that the same is available for operation
- PC9.** If mould is not available collect the mould from tool room
- PC10.** Install and bolt the mould in place and slide the safety door shut
- PC11.** Add the raw material in the machine using material loader or by manual feeding.

Clean the apparatus and the components before executing the process

To be competent, the user/individual on the job must be able to:

- PC12.** Ensure moulds are clean if not clean with soft cotton cloth
- PC13.** Ensure cleaning of the other auxiliaries tools, (if any) before the initiation of the moulding and trimming process

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PC14. Ensure cleaning of the area around the apparatus for any oil, grease, combustible substances etc. so as to prevent any accident

PC15. Ensure availability of the coolant and working of valves to circulate the coolant to cool and solidify plastic

Check materials and apparatus for Operations

To be competent, the user/individual on the job must be able to:

PC16. Understand the raw material like plastics granules, fillers, bonding additives etc. required for executing the activity

Escalations of queries on the given job

To be competent, the user/individual on the job must be able to:

PC17. Refer the queries to supervisor if they cannot be resolved by the operator

PC18. Confirm self - understanding to the supervisor once the query is resolved so that all doubts & queries can be resolved before the actual process execution

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Company code of conduct

KU2. Different types of products manufactured by the company

KU3. Functional processes like Procurement, Store management, inventory management, quality management, incentives, personnel management

KU4. Importance of individual role in the work flow

KU5. Organization culture

KU6. Companys reporting structure

KU7. Companys documentation policy

KU8. General principles of moulding procedure and process knowledge mould loading and unloading procedure

KU9. Types of plastics like thermoplastics and the additives & grades to be used tonnage and capacity of the machine being operated

KU10. Different types of tools and machinery to process the plastic and trim the output

KU11. Various types of cooling systems and their properties

KU12. Geometry and dimension measurement of the product output

KU13. How to perform moulding machine safety check

KU14. Measuring instruments like vernier callipers, micrometres, Thickness Gauge

KU15. Hazards and safety aspects involved in tape production and usage of relevant PPEs

KU16. Safety procedures to be adopted to complete mould removal process

Generic Skills (GS)

User/individual on the job needs to know how to:

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- GS1.** How to be able to read warnings, instructions and other text material on product labels, components etc
- GS2.** How to enter into the history card details of the fault identified in the plastic product manufactured read equipment manuals and process documents to understand the equipment and processes better its
- GS3.** Read instructions especially safety instructions especially symbols while using the equipment in the plant area logs
- GS4.** Discuss task lists, schedules, and work-loads with co-workers
- GS5.** Question internal customers/ Shop floor supervisor appropriately in order to understand the nature of the problem and make a DiagLOis
- GS6.** Avoid using jargon, slang or acronyms when communicating with a supervisor /fellow subordinates etc. Unless it is required
- GS7.** Support supervisor in using specific problem solving techniques and detailing out the problems
- GS8.** Discuss possible solution with the supervisor for problem solving
- GS9.** Make decisions in emergency conditions in case the supervisor is not available (as per the authority matrix defined by the Organization)
- GS10.** Plan and organize the work order and jobs received from the internal customers
- GS11.** Plan and organize the design documents received from internal customers
- GS12.** Organize all process/ equipment manuals so that sorting out
- GS13.** Follow instructions and work on areas of improvement identified
- GS14.** Complete the assigned tasks with minimum supervision
- GS15.** Complete the job defined by the supervisor within the timelines and quality
- GS16.** Use common sense and make judgments during day to day basis
- GS17.** Use reasoning skills to identify and resolve basic problems
- GS18.** Use intuition to detect any potential problems which could arise during operations

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Understand the work and the process requirements</i>	15	17	-	-
PC1. To interact with the supervisor in order to understand the production schedule	3	3	-	-
PC2. To plan the day's production activities based on the supervisor's instructions	3	3	-	-
PC3. To ensure availability of consumables and plastics materials for production in sufficient quantity as per production plan/supervisor instructions	3	3	-	-
PC4. Clearly understanding the does and don'ts of the manufacturing process as defined in SOPs/ Work Instructions or defined by supervisors	2	3	-	-
PC5. Check availability of the personal protective equipments (PPE) like Gloves, Goggles etc.	2	3	-	-
PC6. Understand the molding procedure and process to be adopted for completing the work order from the supervisor by referring the Work Instruction document/ SOP manual.	2	2	-	-
<i>Arrange for the material to be moulded and Mould required for the same</i>	10	9	-	-
PC7. Ensure that the required material is procured from the store before starting the process	2	2	-	-
PC8. Understand the Mould required for executing the required operation and ensure that the same is available for operation	2	2	-	-
PC9. If mould is not available collect the mould from tool room	2	2	-	-
PC10. Install and bolt the mould in place and slide the safety door shut	2	2	-	-
PC11. Add the raw material in the machine using material loader or by manual feeding.	2	1	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Clean the apparatus and the components before executing the process</i>	8.5	5	-	-
PC12. Ensure moulds are clean if not clean with soft cotton cloth	2	1	-	-
PC13. Ensure cleaning of the other auxiliaries tools, (if any) before the initiation of the moulding and trimming process	2.5	1	-	-
PC14. Ensure cleaning of the area around the apparatus for any oil, grease, combustible substances etc. so as to prevent any accident	2.5	1	-	-
PC15. Ensure availability of the coolant and working of valves to circulate the coolant to cool and solidify plastic	1.5	2	-	-
<i>Check materials and apparatus for Operations</i>	1.5	1	-	-
PC16. Understand the raw material like plastics granules, fillers, bonding additives etc. required for executing the activity	1.5	1	-	-
<i>Escalations of queries on the given job</i>	3	2	-	-
PC17. Refer the queries to supervisor if they cannot be resolved by the operator	1.5	1	-	-
PC18. Confirm self - understanding to the supervisor once the query is resolved so that all doubts & queries can be resolved before the actual process execution	1.5	1	-	-
NOS Total	38	34	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	CPC/CAP/N0201
NOS Name	Basic concept, job requirements and basics know how related to the process
Sector	Chemical & PetroChemical
Sub-Sector	
Occupation	Injection Moulding
NSQF Level	3.5
Credits	3
Version	1.0
Last Reviewed Date	26/05/2025
Next Review Date	25/05/2028
NSQC Clearance Date	26/05/2025

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CPC/CAP/N0202: Injection moulding related operations, monitor process parameters and troubleshoot the process/product if any

Description

This NOS unit is about produce good quality moldings in line with the required specifications

Scope

The scope covers the following :

- The Injection Moulding machine operator will be responsible for
- Checking the operations of the equipment
- Feeding the granules as per requirement
- Set up and operate the Injection moulding machine
- Perform visual inspection of the output products
- Achieve productivity, quality and safety standards as per company's norms
- Report problems to supervisor

Elements and Performance Criteria

Check the operations of the equipment used in the molding process

To be competent, the user/individual on the job must be able to:

- PC1.** Check for operation of molding apparatus like hopper, heaters etc. as per the checklist provided
- PC2.** Fix the desired Mould to the injection moulding machine in order to achieve the desired operation as per the Work Instructions/ SOPs
- PC3.** Make modifications in the process parameters (by selecting the right program from the machine control system) if required and ensure alignment with the prescribed standards

Feed the plastic granules in the hopper and conduct a trial product with the setting

To be competent, the user/individual on the job must be able to:

- PC4.** Perform preheating of plastic granules (In case of Engineering plastics)
- PC5.** Ensure that the plastic granules are mixed with additives (if any) before being fed into the hopper
- PC6.** Conduct a test process and produce a sample output as per the required
- PC7.** Ensure that the dimensions of the output product are measured as per the process given in the Work Instructions/ SOP
- PC8.** In case the test product matches the dimensions and quality of the final output, start the production process

Conduct the actual process with final setting as per product approval

To be competent, the user/individual on the job must be able to:

- PC9.** Feed the required operation code in the apparatus for heaters to melt the plastic granules at the predefined temperature

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- PC10.** Enter moulding temperature, volume of plastic and weight settings in the machine as per data sheet
- PC11.** Enter machine and process parameters such as moulding pressure and time as per the data sheet
- PC12.** Add master batch and regrind raw material as per standard composition and mix it well
- PC13.** Check-list procedure to ensure quality of final product

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Departments code of conduct
- KU2.** Different types machines in the company, Its specifications etc.
- KU3.** Department documentation policy
- KU4.** General principles of moulding machine operations, Startup, Shutdown etc.
- KU5.** Process parameters setting, producing good product etc.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Read and interpret engineering drawing and sketches
- GS2.** Read equipment manuals and process documents to understand the equipment and processes better
- GS3.** Read instructions especially safety instructions especially symbols while using the equipment in the plant area
- GS4.** Read internal drawings send by internal customers (other functions within the organization)
- GS5.** Document information from the sketches and engineering drawings
- GS6.** Write log book in terms of output quantity, set up parameters, machine setting parameters and loss details etc.
- GS7.** Prepare draft drawings for the final output product
- GS8.** Write drawings to internal customers on the requirement of moulding plastic, molding apparatus etc.
- GS9.** Note measurements, equipment panel readings for various process parameters in the required reporting formats
- GS10.** How to visualize final product output and hence decide on the key steps to be followed
- GS11.** Safety precautions to be taken for entire extrusion, post extrusion& loom machine setting activities
- GS12.** How to avoid defects in machine operation and final product manufacture
- GS13.** Discuss task lists, schedules, and work-loads with co-workers
- GS14.** Question internal customers/ Moulding shop supervisor appropriately in order to understand the nature of the problem and make a DiagLOis
- GS15.** Detect problems in day to day tasks

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- GS16.** Support supervisor in using specific problem solving techniques and detailing out the problems
- GS17.** Discuss possible solution with the supervisor for problem solving
- GS18.** Make decisions in emergency conditions in case the supervisor is not available(as per the authority matrix defined by the organization)
- GS19.** Plan and organize the work order and jobs received from the internal customers
- GS20.** Plan and organize the design documents received from internal Customers
- GS21.** Organize all process/ equipment manuals so that sorting out information is Fast
- GS22.** Organize apparatus etc. In an orderly manner at proper designated areas Analysis of defects that occur in the final products and Correlate them to Problems with the injection moulding machine
- GS23.** How to combine machine dependent and machine independent settings
- GS24.** How to visualize final product output and hence decide on the key steps to be followed
- GS25.** Follow instructions and work on areas of improvement identified
- GS26.** Complete the assigned tasks with minimum supervision
- GS27.** Complete the job defined by the supervisor within the timelines and Quality
- GS28.** Visualize the final job product after understanding the given drawing/ sketches
- GS29.** Carefully measure the moulding so in terms of the geometrical dimensions so that the final output is as pre the given drawing
- GS30.** Finalize the optimum levels of physical parameters so that the job output meets the prescribed job standards

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Check the operations of the equipment used in the molding process</i>	30	54	-	-
PC1. Check for operation of molding apparatus like hopper, heaters etc. as per the checklist provided	10	18	-	-
PC2. Fix the desired Mould to the injection moulding machine in order to achieve the desired operation as per the Work Instructions/ SOPs	10	18	-	-
PC3. Make modifications in the process parameters (by selecting the right program from the machine control system) if required and ensure alignment with the prescribed standards	10	18	-	-
<i>Feed the plastic granules in the hopper and conduct a trial product with the setting</i>	31	75	-	-
PC4. Perform preheating of plastic granules (In case of Engineering plastics)	5	15	-	-
PC5. Ensure that the plastic granules are mixed with additives (if any) before being fed into the hopper	5	15	-	-
PC6. Conduct a test process and produce a sample output as per the required	5	10	-	-
PC7. Ensure that the dimensions of the output product are measured as per the process given in the Work Instructions/ SOP	10	20	-	-
PC8. In case the test product matches the dimensions and quality of the final output, start the production process	6	15	-	-
<i>Conduct the actual process with final setting as per product approval</i>	25	100	-	-
PC9. Feed the required operation code in the apparatus for heaters to melt the plastic granules at the predefined temperature	5	20	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. Enter moulding temperature, volume of plastic and weight settings in the machine as per data sheet	5	20	-	-
PC11. Enter machine and process parameters such as moulding pressure and time as per the data sheet	5	20	-	-
PC12. Add master batch and regrind raw material as per standard composition and mix it well	5	20	-	-
PC13. Check-list procedure to ensure quality of final product	5	20	-	-
NOS Total	86	229	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	CPC/CAP/N0202
NOS Name	Injection moulding related operations, monitor process parameters and troubleshoot the process/product if any
Sector	Chemical & PetroChemical
Sub-Sector	
Occupation	Injection Moulding
NSQF Level	3.5
Credits	9
Version	1.0
Last Reviewed Date	26/05/2025
Next Review Date	25/05/2028
NSQC Clearance Date	26/05/2025

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CPC/CAP/N0203: Conduct quality checks and inspection of the finished products with reference to the approved product

Description

This NOS unit is about inspecting the finished goods produced for any damages, deformities and further repairing the parts produced so that the damaged/ defective pieces can be corrected and right quality components are supplied to 1. The customer/ end user 2. Internal manufacturing team

Scope

The scope covers the following :

- The Injection moulding operator will be responsible for
- Inspecting the finished components
- keeping records of production and defects
- conducting minor repair/deflashing if any on output parts which can be reworked
- The role holder will interact with maintenance team and material management team

Elements and Performance Criteria

Inspection of finished goods to detect any deviations from the approved product

To be competent, the user/individual on the job must be able to:

PC1. Compare texture, colour, surface properties, hardness and strength etc. with the given approved product

Record log of defective products and discard defective product

To be competent, the user/individual on the job must be able to:

PC2. • Rectify minor defects like dimension variation, thickness variation etc. by controlling process parameters etc and informing operators.

PC3. • Provide first and last output from each batch to the lab for quality check on its composition, properties
• etc.

PC4. Obtain clearance for the entire batch from the lab and submit the operator

PC5. Establish linkage between rejection of output and the pertinent causes for the same (process/ material etc.); Recommend the means for rejection control

Corrective batch process with minor defects

To be competent, the user/individual on the job must be able to:

PC6. Rectify minor defects like dimension variation, thickness variation etc. by control process parameters etc.

PC7. Escalate all issues related to change in surface properties, Tensile strength etc. so that the manufacturing equipment can be reset to achieve the specified output.

Perform Batch Quality Procedure

To be competent, the user/individual on the job must be able to:

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- PC8.** Provide first and last output from each batch to the lab for quality check on its composition, properties etc.
- PC9.** Obtain clearance for the entire batch from the lab

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Relevant standards specified for the manufacturing process
- KU2.** Basic process followed for inspection of the lot
- KU3.** Quality Management policy of the organization
- KU4.** Processes and procedures followed for manufacturing the lot/ prices/ products
- KU5.** Techniques of using measurement instruments like rulers, Vernier calipers, micrometers, weighing scales etc.
- KU6.** Methods to identify quality defects in the lot.
- KU7.** Impact of defects on the overall working of the Injection Moulding machine
- KU8.** Methods used for cutting, finishing which can repair lot (Moulded products) with minor defects
- KU9.** Various quality standards used by the organization

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Note the number of lot with defects which can be repaired to number of lot which will be discarded
- GS2.** Read process and equipment manuals, material data sheets etc. to understand the working of the equipment & material properties
- GS3.** Inform supervisor of any quality related defects arising out of the manufacturing process
- GS4.** Question internal customers/ supervisor appropriately in order to understand the nature of the problem and make a DiagLOis
- GS5.** Plan and organize the work order and jobs received from the supervisor
- GS6.** Organize all process/ equipment manuals so that sorting/ accessing information is easy
- GS7.** Keep fixtures, tools, drawings, Work Instructions, SOP manuals as per the part number, colour codes etc. as defined under the 5S systems
- GS8.** Use common sense and make judgments during day to day basis use reasoning skills to identify and resolve basic problems
- GS9.** Carefully analyze the body part for various assembling defects at every station
- GS10.** Carefully analyze each defect observed during inspection and try to find solution for the defect along with the assembly line operator
- GS11.** Identify defective parts in the manufacturing line by comparing manufactured (lot/extrudate) with the work standard
- GS12.** Link the defect observed with the overall impact on the performance of the (lot/extrudate)

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Inspection of finished goods to detect any deviations from the approved product</i>	4	10	-	-
PC1. Compare texture, colour, surface properties, hardness and strength etc. with the given approved product	4	10	-	-
<i>Record log of defective products and discard defective product</i>	12	38	-	-
PC2. • Rectify minor defects like dimension variation, thickness variation etc. by controlling process • parameters etc and informing operators.	3	10	-	-
PC3. • Provide first and last output from each batch to the lab for quality check on its composition, properties • etc.	3	10	-	-
PC4. Obtain clearance for the entire batch from the lab and submit the operator	3	10	-	-
PC5. Establish linkage between rejection of output and the pertinent causes for the same (process/ material etc.); Recommend the means for rejection control	3	8	-	-
<i>Corrective batch process with minor defects</i>	6	18	-	-
PC6. Rectify minor defects like dimension variation, thickness variation etc. by control process parameters etc.	3	8	-	-
PC7. Escalate all issues related to change in surface properties, Tensile strength etc. so that the manufacturing equipment can be reset to achieve the specified output.	3	10	-	-
<i>Perform Batch Quality Procedure</i>	4	9	-	-
PC8. Provide first and last output from each batch to the lab for quality check on its composition, properties etc.	2	5	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC9. Obtain clearance for the entire batch from the lab	2	4	-	-
NOS Total	26	75	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	CPC/CAP/N0203
NOS Name	Conduct quality checks and inspection of the finished products with reference to the approved product
Sector	Chemical & PetroChemical
Sub-Sector	
Occupation	Injection Moulding
NSQF Level	3.5
Credits	3
Version	1.0
Last Reviewed Date	26/05/2025
Next Review Date	25/05/2028
NSQC Clearance Date	26/05/2025

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CPC/CAP/N0204: Entrepreneurship in Injection Moulding

Description

This NOS unit is about entrepreneurship in Plastic Injection Moulding

Scope

The scope covers the following :

- This unit/task covers the following:
- Market Information Management
- Client Relation Management
- Marketing

Elements and Performance Criteria

Injection moulding Economics and Finances

To be competent, the user/individual on the job must be able to:

- PC1.** Planning and Budgeting with reference to various components of Injection Moulding
- PC2.** Keep books of accounts and various transactions
- PC3.** Arrange for financial assistance from various quarters in the light of various schemes available in setup for Injection Moulding

Market Information Management

To be competent, the user/individual on the job must be able to:

- PC4.** Ascertain the prices of various inputs and products from the market
- PC5.** Assess the influence of various quality parameters of products on the product pricing

Client Relation Management

To be competent, the user/individual on the job must be able to:

- PC6.** Establish cordial relations with various clients for the benefit of industry
- PC7.** Assess the needs and requirement of the clients and assess one's own unique selling proposition
- PC8.** Extract critical market information that is otherwise not in the public domain

Marketing

To be competent, the user/individual on the job must be able to:

- PC9.** Choose appropriate buyer in a given situation of market parameters
- PC10.** Identify best ways of attracting market price for one's produce
- PC11.** Ensure quality before & during the sale activity to ensure good returns

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

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- KU1.** Basic steps of Injection Moulding planning and budgeting
- KU2.** Basic principles of keeping books of accounts
- KU3.** Various Government and other schemes / products / offers available for startup and support of Injection Moulding
- KU4.** Different players selling various injection moulded products and their prices
- KU5.** Different players buying injection moulded products & their prices
- KU6.** Various methods of updating oneself with market information such as mobile, Internet etc.
- KU7.** Usage, contact with key informants, tie up government agencies etc.
- KU8.** Needs and options available with various clients
- KU9.** Advantages and disadvantages of doing business with each one of the clients
- KU10.** The quality parameters of injection moulded products and their market prices
- KU11.** Pricing mechanism of various buyers of injection moulded products
- KU12.** Costing of various logistic arrangements towards the sale injection moulded products at different markets and consumer points

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Mention the data which are required for record keeping purpose
- GS2.** Report problems to the appropriate personnel in a timely manner
- GS3.** Write descriptions and details about incidents in reports
- GS4.** Keep abreast with the latest knowledge by reading brochures, pamphlets and product information sheets
- GS5.** Read instruction manuals for hand tool and equipments
- GS6.** Read instructions on work orders and procedures
- GS7.** Discuss task lists, schedules, and work-loads with co-workers
- GS8.** Question customers appropriately in order to understand the nature of the problem and make a DiagLOis
- GS9.** Give clear instructions to customers
- GS10.** Keep customers informed about progress
- GS11.** Avoid using jargon, slang or acronyms when communicating with a customer, unless it is required
- GS12.** Make decisions pertaining to the concerned area of work
- GS13.** Plan and organize service feedback files/documents
- GS14.** Manage relationships with customers who may be stressed, frustrated, confused, or angry
- GS15.** Build customer relationships and use customer centric approach
- GS16.** Think through the problem, evaluate the possible solution(s) and suggest an optimum /best possible solution(s)
- GS17.** Deal with clients lacking the technical background to solve the problem on their own
- GS18.** Identify immediate or temporary solutions to resolve delays



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- GS19.** Use the existing data to arrive at specific data points
- GS20.** Use the existing data points for improving the defect resolution time
- GS21.** Use the existing data points to generate required reports for business
- GS22.** Apply, analyze, and evaluate the information gathered from observation, experience, reasoning, or communication, as a guide to thought and action

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Injection moulding Economics and Finances</i>	6	1.5	-	-
PC1. Planning and Budgeting with reference to various components of Injection Moulding	2	0.5	-	-
PC2. Keep books of accounts and various transactions	2	0.5	-	-
PC3. Arrange for financial assistance from various quarters in the light of various schemes available in setup for Injection Moulding	2	0.5	-	-
<i>Market Information Management</i>	4	1.5	-	-
PC4. Ascertain the prices of various inputs and products from the market	2	0.5	-	-
PC5. Assess the influence of various quality parameters of products on the product pricing	2	1	-	-
<i>Client Relation Management</i>	6	2	-	-
PC6. Establish cordial relations with various clients for the benefit of industry	2	0.5	-	-
PC7. Assess the needs and requirement of the clients and assess one's own unique selling proposition	2	0.5	-	-
PC8. Extract critical market information that is otherwise not in the public domain	2	1	-	-
<i>Marketing</i>	6	3	-	-
PC9. Choose appropriate buyer in a given situation of market parameters	2	1	-	-
PC10. Identify best ways of attracting market price for one's produce	2	1	-	-
PC11. Ensure quality before & during the sale activity to ensure good returns	2	1	-	-
NOS Total	22	8	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	CPC/CAP/N0204
NOS Name	Entrepreneurship in Injection Moulding
Sector	Chemical & PetroChemical
Sub-Sector	
Occupation	Injection Moulding
NSQF Level	3.5
Credits	1
Version	1.0
Last Reviewed Date	26/05/2025
Next Review Date	25/05/2028
NSQC Clearance Date	26/05/2025

Qualification Pack

CPC/CAP/N0401: Basic health and safety practices at the workplace, 5S

Description

This unit is about creating a Safe and Healthy work place, adhering to the safety guidelines in the working area, following practices which are not impacting the environment in a negative manner. This NOS is about ensuring all 5S activities both at the shop floor and the office area to facilitate increase in work productivity. It covers knowledge of fire safety, common first aid applications, safe practice and emergency applications.

Scope

The scope covers the following :

- The role holder will be responsible for
- Identifying and reporting of risks
- Creating and sustaining a safe, clean and environment friendly Work place.
- Ensure sorting, streamlining & organizing, storage and documentation, cleaning, standardization and sustenance across the plant and office premises of the organization

Elements and Performance Criteria

Health and safety

To be competent, the user/individual on the job must be able to:

- PC1.** Wear protective clothing/equipment for specific tasks and work conditions
- PC2.** Carry out safe working practices while dealing with hazards to ensure the safety of Self and others.
- PC3.** Apply good housekeeping practices at all times

Fire safety

To be competent, the user/individual on the job must be able to:

- PC4.** Use the various appropriate fire extinguishers on different types of fires correctly
- PC5.** Demonstrate rescue techniques applied during fire hazard, demonstrate good housekeeping in order to prevent fire hazards, demonstrate the correct use of a fire extinguisher.

Identify and report the risks identified

To be competent, the user/individual on the job must be able to:

- PC6.** Identify activities which can cause potential injury through sharp objects, burns, fall, electricity, gas leakages, radiation, poisonous fumes, chemicals, loud noise, and Identify areas in the plant which are potentially hazardous/ unhygienic in nature. Conduct regular checks with support of the maintenance team on machine health to identify potential hazards due to wear and tear of machine.
- PC7.** Inform the concerned authorities on the potential risks identified in the processes, workplace area/ layout, materials used etc, Inform the concerned authorities about machine breakdowns, damages which can potentially harm man/ machine during operations.
- PC8.** Create awareness amongst other by sharing information on the identified risks.

Ensure sorting

Qualification Pack

To be competent, the user/individual on the job must be able to:

- PC9.** Follow the sorting process and check that the tools, fixtures & jigs that are lying on workstations are the ones in use and unnecessary items are not cluttering the workbenches or work surfaces.
- PC10.** Ensure segregation of waste in hazardous/ non Hazardous waste as per the sorting work instructions
- PC11.** Follow the technique of waste disposal and waste storage in the proper bins as per SOP
- PC12.** Segregate the items which are labeled as red tag items for the process area and keep them in the correct places
- PC13.** Sort the tools/ equipment/ fasteners/ spare parts as per specifications/ utility into proper trays, cabinets, lockers as mentioned in the 5S guidelines/ work instructions
- PC14.** Ensure that areas of material storage areas are not overflowing
- PC15.** Properly stack the various types of boxes and containers as per the size/ utility to avoid any fall of items/ breakage and also enable easy sorting when required
- PC16.** Return the extra material and tools to the designated sections and make sure that no additional material/ tool is lying near the work area
- PC17.** Follow the floor markings/ area markings used for demarcating the various sections in the plant as per the prescribed instructions and standards

Ensure proper documentation and storage(organizing, streamlining)

To be competent, the user/individual on the job must be able to:

- PC18.** Follow the proper labelling mechanism of instruments/ boxes/ containers and maintaining reference files/ documents with the codes and the lists
- PC19.** Check that the items in the respective areas have been identified as broken or damaged
- PC20.** Follow the given instructions and check for labelling of fluids, oils, lubricants, solvents, chemicals etc. and proper storage of the same. To avoid spillage, leakage, fire etc.
- PC21.** Make sure that all material and tools are stored in the designated places and in the manner indicated in the 5S instructions

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Relevant standards, procedures and policies related to Health, Safety and Environment followed in the company
- KU2.** Emergency handling procedures & hierarchy for escalation
- KU3.** Basic knowledge of Safety procedures(firefighting, first aid) within the organization
- KU4.** Basic knowledge of various types of PPEs and their usage
- KU5.** Basic knowledge of risks/hazards associated with each occupation in the organization
- KU6.** Knowledge of personal hygiene and how an individual can contribute towards creating a highly safe and clean working environment
- KU7.** Meaning of hazards and risks
- KU8.** Health and safety hazards commonly present in the work environment and related precautions

Qualification Pack

- KU9.** Possible causes of risk, hazard or accident in the workplace and why risk and/or accidents are possible
- KU10.** Possible causes of risk and accident (due to oil leakage)
- KU11.** Safe working practices when working with tools and machines
- KU12.** Safe working practices while working at various hazardous sites
- KU13.** Where to find all the general health and safety equipment in the workplace
- KU14.** Various dangers associated with the use of electrical equipment
- KU15.** Preventative and remedial actions to be taken in the case of exposure to toxic materials
- KU16.** Importance of using protective clothing/equipment while working
- KU17.** Precautionary activities to prevent the fire accident
- KU18.** Various causes of fire
- KU19.** Techniques of using the different fire extinguishers
- KU20.** Different methods of extinguishing fire
- KU21.** Different materials used for extinguishing fire
- KU22.** Rescue techniques applied during a fire hazard
- KU23.** Various types of safety signs and what they mean
- KU24.** Appropriate basic first aid treatment relevant to the condition eg. shock, electrical shock, bleeding, breaks to bones, minor burns, resuscitation, poisoning, eye injuries
- KU25.** Content of written accident report
- KU26.** Potential injuries and ill health associated with incorrect manual handling
- KU27.** Safe lifting and carrying practices
- KU28.** Personal safety, health and dignity issues relating to the movement of a person by others
- KU29.** Potential impact to a person who is moved incorrectly
- KU30.** Have basic knowledge of 5S procedures
- KU31.** Know various types 5s practices followed in various areas
- KU32.** Understand the 5S checklists provided in the department/ team
- KU33.** Have skills to identify useful & non useful items
- KU34.** Have knowledge of labels , signs & colors used as indicators
- KU35.** Have knowledge on how to sort and store various types of tools, equipment, material etc.
- KU36.** Know , how to identify various types of waste products
- KU37.** Understand the impact of waste/ dirt/ dust/unwanted substances on the process/ environment/ machinery/ human body.
- KU38.** Have knowledge of best ways of cleaning & waste disposal

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Write basic level notes and observations
- GS2.** Read safety instructions put up across the plant premises

Qualification Pack

- GS3.** Read safety precautions mentioned in equipment manuals and panels to understand the potential risks associated
- GS4.** Effectively communicate information to team members
- GS5.** Inform employees in the plant and concerned functions about events, incidents & potential risks observed related to Safety, Health and Environment.
- GS6.** Question operator/ supervisor in order to understand the safety related issues
- GS7.** Attentively listen with full attention and comprehend the information given by the speaker during safety drills and training programs
- GS8.** Plan and organize the work order and jobs received from the internal customers
- GS9.** Plan and organize the design documents received from internal customers
- GS10.** Organize all process/ equipment manuals so that sorting out information is Fast
- GS11.** Use common sense and make judgments during day to day basis
- GS12.** Use reasoning skills to identify and resolve basic problems
- GS13.** Use intuition to detect any potential problems which could arise during operations
- GS14.** Follow instructions and work on areas of improvement identified
- GS15.** Complete the assigned tasks with minimum supervision
- GS16.** Complete the job defined by the supervisor within the timelines and quality norms

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Health and safety</i>	1.5	6	-	-
PC1. Wear protective clothing/equipment for specific tasks and work conditions	0.5	2	-	-
PC2. Carry out safe working practices while dealing with hazards to ensure the safety of Self and others.	0.5	2	-	-
PC3. Apply good housekeeping practices at all times	0.5	2	-	-
<i>Fire safety</i>	1	4	-	-
PC4. Use the various appropriate fire extinguishers on different types of fires correctly	0.5	2	-	-
PC5. Demonstrate rescue techniques applied during fire hazard, demonstrate good housekeeping in order to prevent fire hazards, demonstrate the correct use of a fire extinguisher.	0.5	2	-	-
<i>Identify and report the risks identified</i>	1.5	6	-	-
PC6. Identify activities which can cause potential injury through sharp objects, burns, fall, electricity, gas leakages, radiation, poisonous fumes, chemicals, loud noise, and Identify areas in the plant which are potentially hazardous/ unhygienic in nature. Conduct regular checks with support of the maintenance team on machine health to identify potential hazards due to wear and tear of machine.	0.5	2	-	-
PC7. Inform the concerned authorities on the potential risks identified in the processes, workplace area/ layout, materials used etc, Inform the concerned authorities about machine breakdowns, damages which can potentially harm man/ machine during operations.	0.5	2	-	-
PC8. Create awareness amongst other by sharing information on the identified risks.	0.5	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Ensure sorting</i>	4	10	-	-
PC9. Follow the sorting process and check that the tools, fixtures & jigs that are lying on workstations are the ones in use and unnecessary items are not cluttering the workbenches or work surfaces.	0.5	2	-	-
PC10. Ensure segregation of waste in hazardous/ non Hazardous waste as per the sorting work instructions	0.5	2	-	-
PC11. Follow the technique of waste disposal and waste storage in the proper bins as per SOP	0.5	1	-	-
PC12. Segregate the items which are labeled as red tag items for the process area and keep them in the correct places	0.5	1	-	-
PC13. Sort the tools/ equipment/ fasteners/ spare parts as per specifications/ utility into proper trays, cabinets, lockers as mentioned in the 5S guidelines/ work instructions	0.5	1	-	-
PC14. Ensure that areas of material storage areas are not overflowing	0.25	0.5	-	-
PC15. Properly stack the various types of boxes and containers as per the size/ utility to avoid any fall of items/ breakage and also enable easy sorting when required	0.25	0.5	-	-
PC16. Return the extra material and tools to the designated sections and make sure that no additional material/ tool is lying near the work area	0.5	1	-	-
PC17. Follow the floor markings/ area markings used for demarcating the various sections in the plant as per the prescribed instructions and standards	0.5	1	-	-
<i>Ensure proper documentation and storage(organizing, streamlining)</i>	2	4	-	-
PC18. Follow the proper labelling mechanism of instruments/ boxes/ containers and maintaining reference files/ documents with the codes and the lists	0.5	1	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC19. Check that the items in the respective areas have been identified as broken or damaged	0.5	1	-	-
PC20. Follow the given instructions and check for labelling of fluids, oils, lubricants, solvents, chemicals etc. and proper storage of the same. To avoid spillage, leakage, fire etc.	0.5	1	-	-
PC21. Make sure that all material and tools are stored in the designated places and in the manner indicated in the 5S instructions	0.5	1	-	-
NOS Total	10	30	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	CPC/CAP/N0401
NOS Name	Basic health and safety practices at the workplace, 5S
Sector	Chemical & PetroChemical
Sub-Sector	
Occupation	Generic
NSQF Level	3.5
Credits	1
Version	1.0
Last Reviewed Date	26/05/2025
Next Review Date	25/05/2028
NSQC Clearance Date	26/05/2025



Qualification Pack

CPC/CAP/N0402: Basics of computer and data entry in MS OFFICE/office Open source suite Software

Description

This NOS unit is about the Data Entry Operation for Injection Moulding like entering, updating and maintain Job work related data the computer systems having MS Office software

Scope

The scope covers the following :

- This unit / task covers the Enter, update and maintain data in MS Office system

Elements and Performance Criteria

Enter, update and maintain data

To be competent, the user/individual on the job must be able to:

- PC1.** Fill and process mandated forms for receiving, processing, or tracking data, enter data from source documents (such as trial report, process sheet etc.) in to Computer application having MS OFFICE software.
- PC2.** Scan source documents in accordance with specific instructions.
- PC3.** Verify data entered with source documents, checks for compliance and corrects all typographical errors and missing or repeated data.
- PC4.** Maintain files of source documents or other information related to data entered.
- PC5.** Investigate and confirm data that is unclear before entering, generate reports of data entry, store completed work in designated locations and perform backup operations.
- PC6.** Update database information to reflect most current source information
- PC7.** Assist in the filing and storage of security and back up data files
- PC8.** Respond to requests for information and access relevant files

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** The data management applications/tools used by the company
- KU2.** Data entry protocol
- KU3.** Data integrity and security policies of the company
- KU4.** Approved methods for carrying document control and archiving
- KU5.** Basic understanding computers and its terminology
- KU6.** Work on different software needed for report writing including MS office suit or open source office

Generic Skills (GS)



Qualification Pack

User/individual on the job needs to know how to:

- GS1.** Efficiently enter data into computer applications
- GS2.** Prepare legible reports
- GS3.** Read and understand manuals, SOPs, instructions, memos, reports, job cards etc.
- GS4.** Communicate effectively with the team members and supervisors
- GS5.** Apply basic logic to identify data errors
- GS6.** Pay attention to details
- GS7.** Plan assigned tasks within timeline and as per priority order specified
- GS8.** Identify process improvements

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Enter, update and maintain data</i>	8	14	-	-
PC1. Fill and process mandated forms for receiving, processing, or tracking data, enter data from source documents (such as trial report, process sheet etc.) in to Computer application having MS OFFICE software.	1	2	-	-
PC2. Scan source documents in accordance with specific instructions.	1	2	-	-
PC3. Verify data entered with source documents, checks for compliance and corrects all typographical errors and missing or repeated data.	1	2	-	-
PC4. Maintain files of source documents or other information related to data entered.	1	2	-	-
PC5. Investigate and confirm data that is unclear before entering, generate reports of data entry, store completed work in designated locations and perform backup operations.	1	2	-	-
PC6. Update database information to reflect most current source information	1	1	-	-
PC7. Assist in the filing and storage of security and back up data files	1	2	-	-
PC8. Respond to requests for information and access relevant files	1	1	-	-
NOS Total	8	14	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	CPC/CAP/N0402
NOS Name	Basics of computer and data entry in MS OFFICE/office Open source suite Software
Sector	Chemical & PetroChemical
Sub-Sector	
Occupation	Generic
NSQF Level	3.5
Credits	1
Version	1.0
Last Reviewed Date	26/05/2025
Next Review Date	25/05/2028
NSQC Clearance Date	26/05/2025

Qualification Pack

DGT/VSQ/N0101: Employability Skills (30 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

PC1. understand the significance of employability skills in meeting the job requirements

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.

Basic English Skills

To be competent, the user/individual on the job must be able to:

PC4. speak with others using some basic English phrases or sentences

Communication Skills

To be competent, the user/individual on the job must be able to:

PC5. follow good manners while communicating with others

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PC6. work with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

PC7. communicate and behave appropriately with all genders and PwD

PC8. report any issues related to sexual harassment

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

PC9. use various financial products and services safely and securely

PC10. calculate income, expenses, savings etc.

PC11. approach the concerned authorities for any exploitation as per legal rights and laws

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

PC12. operate digital devices and use its features and applications securely and safely

PC13. use internet and social media platforms securely and safely

Entrepreneurship

To be competent, the user/individual on the job must be able to:

PC14. identify and assess opportunities for potential business

PC15. identify sources for arranging money and associated financial and legal challenges

Customer Service

To be competent, the user/individual on the job must be able to:

PC16. identify different types of customers

PC17. identify customer needs and address them appropriately

PC18. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC19. create a basic biodata

PC20. search for suitable jobs and apply

PC21. identify and register apprenticeship opportunities as per requirement

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use basic spoken English language

KU6. Do and dont of effective communication

KU7. inclusivity and its importance



Qualification Pack

- KU8.** different types of disabilities and appropriate communication and behaviour towards PwD
- KU9.** different types of financial products and services
- KU10.** how to compute income and expenses
- KU11.** importance of maintaining safety and security in financial transactions
- KU12.** different legal rights and laws
- KU13.** how to operate digital devices and applications safely and securely
- KU14.** ways to identify business opportunities
- KU15.** types of customers and their needs
- KU16.** how to apply for a job and prepare for an interview
- KU17.** apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate effectively using appropriate language
- GS2.** behave politely and appropriately with all
- GS3.** perform basic calculations
- GS4.** solve problems effectively
- GS5.** be careful and attentive at work
- GS6.** use time effectively
- GS7.** maintain hygiene and sanitisation to avoid infection

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. understand the significance of employability skills in meeting the job requirements	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	1	3	-	-
PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC4. speak with others using some basic English phrases or sentences	-	-	-	-
<i>Communication Skills</i>	1	1	-	-
PC5. follow good manners while communicating with others	-	-	-	-
PC6. work with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	1	-	-
PC7. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC8. report any issues related to sexual harassment	-	-	-	-
<i>Financial and Legal Literacy</i>	3	4	-	-
PC9. use various financial products and services safely and securely	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. calculate income, expenses, savings etc.	-	-	-	-
PC11. approach the concerned authorities for any exploitation as per legal rights and laws	-	-	-	-
<i>Essential Digital Skills</i>	4	6	-	-
PC12. operate digital devices and use its features and applications securely and safely	-	-	-	-
PC13. use internet and social media platforms securely and safely	-	-	-	-
<i>Entrepreneurship</i>	3	5	-	-
PC14. identify and assess opportunities for potential business	-	-	-	-
PC15. identify sources for arranging money and associated financial and legal challenges	-	-	-	-
<i>Customer Service</i>	2	2	-	-
PC16. identify different types of customers	-	-	-	-
PC17. identify customer needs and address them appropriately	-	-	-	-
PC18. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	1	3	-	-
PC19. create a basic biodata	-	-	-	-
PC20. search for suitable jobs and apply	-	-	-	-
PC21. identify and register apprenticeship opportunities as per requirement	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0101
NOS Name	Employability Skills (30 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	2
Credits	1
Version	1.0
Last Reviewed Date	12/03/2026
Next Review Date	12/03/2029
NSQC Clearance Date	12/03/2026

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

This section includes the processes involved in identifying, gathering, and interpreting information to evaluate the Candidate on the required competencies of

the program.

Mention the detailed assessment strategy in the provided template.

1. Assessment System Overview:

- Batches are assigned to Training Assessment Wing (TAW), CIPET HO for planning of assessment
- Training Centers request TAW for Assessment and Certification of Trainees
- TAW identifies suitable assessor and nominates the assessor to the respective Training Centre
- TAW monitors the assessment process



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- Training Centers maintain necessary records

2. Testing Environment:

- Check the Assessment location, date and time
- If the batch size is more than 30, then there should be 02 Assessors in a day (or) 01 assessor in 2 days
- Check that the allotted time to the candidates to complete the Theory & Practical Assessment

3. Assessment Quality Assurance levels/Framework:

- Question bank / Question Paper is prepared by the Subject Matter Experts (SME) / Assessor
- Questions are mapped to the specified assessment criteria
- Certified Assessor & Trainer will be engaged in the process

4. Types of evidence or evidence-gathering protocol:

- Date / Time recorded for the reporting of the assessor from assessment location
- Assessment batch - Group Photo of Trainees along with Assessor

5. Method of verification or validation:

- Surprise visit to the assessment location
- Virtual meet with the Assessor / Trainees

6. Method for assessment documentation, archiving, and access

- Hard copies of the documents are stored, soft copies of assessment evidences are stored in Email for future correspondence

Minimum Aggregate Passing % at QP Level : 50

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

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National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
CPC/CAP/N0201.Basic concept, job requirements and basics know how related to the process	38	34	0	0	72	15
CPC/CAP/N0202.Injection moulding related operations, monitor process parameters and troubleshoot the process/product if any	86	229	0	0	315	25
CPC/CAP/N0203.Conduct quality checks and inspection of the finished products with reference to the approved product	26	75	0	0	101	15
CPC/CAP/N0204.Entrepreneurship in Injection Moulding	22	8	0	0	30	15
CPC/CAP/N0401.Basic health and safety practices at the workplace, 5S	10	30	-	-	40	10
CPC/CAP/N0402.Basics of computer and data entry in MS OFFICE/office Open source suite Software	8	14	-	-	22	10
DGT/VSQ/N0101.Employability Skills (30 Hours)	20	30	-	-	50	10
Total	210	420	-	-	630	100



Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

Qualification Pack

Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.